Business Statistics

Mean, Median, and Mode

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Agenda

- Mean, Median and Mode
- Geometric Mean
- Harmonic Mean

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Mean, Median, and Mode

- Mean (Arithmetic Mean)

$$\bar{x} = \frac{Sum\ of\ observations}{Number\ of\ observations}$$

- Median

$\frac{n+1}{2}th$ item when n items arranged in order

- Mode:The most commonly occurring value

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Mean, Median, and Mode

Example

x
28
45
91
68
47
28
37
78

Mean=?

Median= ?

Mode=?

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Mean, Median, and Mode

Example

X
28
45
91
68
47
28
37
78

Mean=52.75

Median= 46

Mode=28

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Geometric Mean

- Another measure of central tendency
- Uses product of values

$$G.M. = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}} = (x_1 x_2 x_3 \dots x_n)^{1/n}$$

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Geometric Mean

Example

X
28
45
91
68
47
28
37
78

$$G.M. = (28 \times 45 \times 91 \times \cdots \times 78)^{1/8}$$
$$= 48.3$$

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Geometric Mean

Geometric Mean is more stable than the arithmetic mean- small or large observations do not cause large fluctuations

The Geometric Mean will be zero if a single value is equal to zero

Applications: Growth Rates, Financial Index, etc.

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Harmonic Mean

$$H.M. = \frac{n}{\sum_{i=1}^{n}\left(\frac{1}{x_i}\right)} = \frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\cdots\frac{1}{x_n}}$$

It is reciprocal of the average (arithmetic mean) of the reciprocals of the observations

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Harmonic Mean

Example

X
28
45
91
68
47
28
37
78

$$H.M. = \frac{8}{\frac{1}{28} + \frac{1}{45} + \frac{1}{91} + \cdots \frac{1}{78}}$$

$$= 44.33$$

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Harmonic Mean

Harmonic Mean is more stable than the arithmetic mean- small or large observations do not cause large fluctuations

Difficult to interpret and not easily computable

Applications: Used for averaging ratios, e.g., Price-Earnings ratio in finance

$$A.M. \geq G.M. \geq H.M.$$

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Summary

- Mean, Median, Mode

- Geometric Mean

$$G.M. = (x_1 x_2 x_3 \dots x_n)^{1/n}$$

- Harmonic Mean

$$H.M. = \frac{n}{\sum_{i=1}^{n}\left(\frac{1}{x_i}\right)} = \frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\cdots\frac{1}{x_n}}$$

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